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University of Ruhuna

Mini Project Proposal

Image Processing and Computer Vision

EE7204 / EC7205

Project Title

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1. Problem Statement

Ground Glass Opacities (GGOs) are faint cloudy areas seen in lung CT scans. They can be early signs of diseases such as pneumonia, lung infections, or even cancer. Finding these areas by hand is difficult and takes a lot of time because GGOs have low contrast and spread out gradually in the lungs. Different doctors may also interpret them differently, which can affect diagnosis accuracy.

Traditional methods do not have clear and standard ways to measure GGOs, Because of this, it is very hard to accurately track how a disease changes over time or to compare scans taken before and after treatment. medical images stored in DICOM format are often handled inefficiently, windowing settings (brightness and contrast) vary between hospitals, and many steps are still done manually instead of automatically. These issues make it difficult to interpret CT scans in a consistent and standardized way.

2. Objectives

- Improve the quality of the chest CT scans by removing the noise, normalizing the image intensity values, and converting the data into a format suitable for processing.
- To accurately segment lung regions using HU-based thresholding, morphological operations, and edge filling techniques while removing background artifacts.
- To detect Ground Glass Opacity regions within the segmented lung area using HU based thresholding criteria.
- Test the efficiency of the proposed image processing workflow on a publicly available chest CT dataset.
- Evaluate the effectiveness of the proposed image processing pipeline using a publicly available chest CT dataset.
- Design and show a simple prototype application which focuses on clear image visualization and the ease of interpretation of the results.

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3. Literature Review

Study 1: Texture based adaptive multiple feature method for pulmonary parenchyma evaluation [1]

Introduced an adaptive multiple feature method (AMFM) using multiple statistical and fractal texture features extracted from CT images to quantify emphysema and related parenchymal abnormalities. The approach divided slices into regions (ventral to dorsal) and applied texture analysis to discriminate normal from abnormal tissue, outperforming simpler mean lung density or histogram percentile methods. It achieved high accuracy (up to 100% globally, ~98% regionally) in separating normal and emphysematous areas. Strengths included objective, quantitative assessment without heavy reliance on fixed thresholds and good regional discrimination. Limitations were its focus on emphysema rather than GGO specifically, computational demands for texture computation at the time, and sensitivity to slice orientation or scanner differences. Relation to our project: We extend intensity and texture concepts to GGO specific ranges (-700 to -500 HU) with added morphological refinement and 3D reconstruction for better spatial context, while keeping the pipeline computationally efficient and interpretable.

Study 2: Region growing and morphological operations for abnormality isolation in CT [2]

Approaches in this era used seeded region growing within initial lung masks (often from thresholding) to isolate ground glass attenuation areas, combined with morphological opening and closing to smooth boundaries and remove noise. These methods correlated CT patterns like ground glass opacification with functional abnormalities in conditions such as hypersensitivity pneumonitis. Strengths lay in improved handling of fuzzy edges and noise compared to pure thresholding, plus integration of local context for better abnormality delineation. Limitations included parameter sensitivity (seed selection, growth criteria) leading to variability across scans, reduced performance on diffuse or faint GGO, and limited 3D volumetric processing. Relation to our project: Our pipeline incorporates connected component analysis post thresholding and advanced morphological operations (erosion, dilation, closing) for robust lung and GGO mask refinement, addressing initialization issues through adaptive HU windows and adding true 3D volume rendering for enhanced visualization.

Study 3: Hybrid lung segmentation with error detection and morphological refinement [3]

Hybrid methods combined rule based segmentation (e.g., thresholding and edge detection) with pixel classification or morphological post processing to segment lung fields, often in chest radiographs but adaptable to CT. These included error detection mechanisms to switch strategies when initial segmentation failed. Strengths were robustness to variations

(noise, pathology) and high accuracy ($\sim 94\text{--}97\%$) through complementary techniques. Limitations involved complexity in hybrid rules, potential for residual errors in severe abnormalities, and less emphasis on GGO specific sub segmentation or interactive 3D views. Relation to our project: We adopt a similar hybrid philosophy starting with HU thresholding for lung extraction, followed by connected components and morphology but extend it to full volumetric 3D processing with GGO candidate highlighting and interactive overlays, prioritizing transparency over classification-heavy hybrids.

Study 4: Markov random field-based segmentation for GGO in lung CT [4]

Probabilistic frameworks using Markov random fields modeled pixel relationships after initial thresholding (e.g., -374 HU or adjusted) and morphological lung boundary refinement (median filtering, opening/closing, erosion). Energy minimization labeled GGO regions with high sensitivity/specificity ($\sim 87\text{--}94\%$). Strengths included effective handling of fuzzy boundaries and noise reduction via probabilistic context. Limitations were computational cost for optimization, need for parameter tuning, and challenges with very subtle or heterogeneous GGO patterns. Relation to our project: While we avoid probabilistic optimization for speed and simplicity, we incorporate similar preprocessing (noise filters, morphology) and HU-based candidate extraction, enhanced by local texture variance checks and interactive 3D pseudo-color rendering to improve clinical interpretability without added complexity.

Study 5: Threshold-based GGO candidate detection with density masks [5]

Density mask techniques applied fixed HU thresholds (e.g., -750 to -300 HU) to quantify opacities, sometimes compared to neural networks for sensitivity gains. These methods segmented GGO areas post-lung isolation and reported high sensitivity but lower specificity due to false positives. Strengths were extreme efficiency and direct radiological alignment (HU-based). Limitations included noise susceptibility, partial volume effects, and poor specificity without refinement, often detecting $17\%+$ extraneous areas. Relation to our project: We use targeted HU ranges (-700 to -500) for GGO but add morphological false-positive reduction and texture discrimination, plus full 3D interactive visualization with overlays addressing specificity and usability gaps in basic density approaches.

Study 6: Semi-automated GGO segmentation with radiomics and threshold tools [6]

Semi-automated pipelines used threshold tools (e.g., -1350 to -700 HU) in software like 3D Slicer, followed by manual correction and morphological/logical operations for lung/GGO ROIs. These supported radiomics feature extraction for differentiation (e.g., COVID-19 vs. other diseases). Strengths included high control and integration with quantification tools. Limitations were partial automation, user dependency, and less

focus on fully interactive 3D prototypes. Relation to our project: Our fully automated yet editable pipeline (with annotation/measurement tools) builds on threshold foundations but emphasizes end-to-end interactivity, 3D rendering, and visual explainability without requiring manual ROI tweaks for initial results.

Gap Analysis and Project Contributions

Existing works excel in foundational thresholding, morphological refinement, and texture analysis for lung/GGO tasks but often lack seamless 3D interactive visualization, color-coded opacity highlighting, and user-friendly annotation/measurement in one tool. Many remain 2D-focused or parameter-sensitive without strong emphasis on reproducibility and clinician control. Our project integrates these classical CV/IP elements ($> 65\%$ focus) into a comprehensive prototype: DICOM preprocessing, adaptive lung segmentation, precise GGO detection with texture/morphology filters, and interactive 3D volume rendering with overlays. This advances interpretability, accessibility, and practical utility for monitoring subtle abnormalities on public datasets.

4. Methodology

The project will consist of 4 main steps:

1. Data Collection and loading
2. Data Preprocessing
3. Lung segmentation
4. GGO visualization and Evaluation

The overall methodology is illustrated in figure 1:

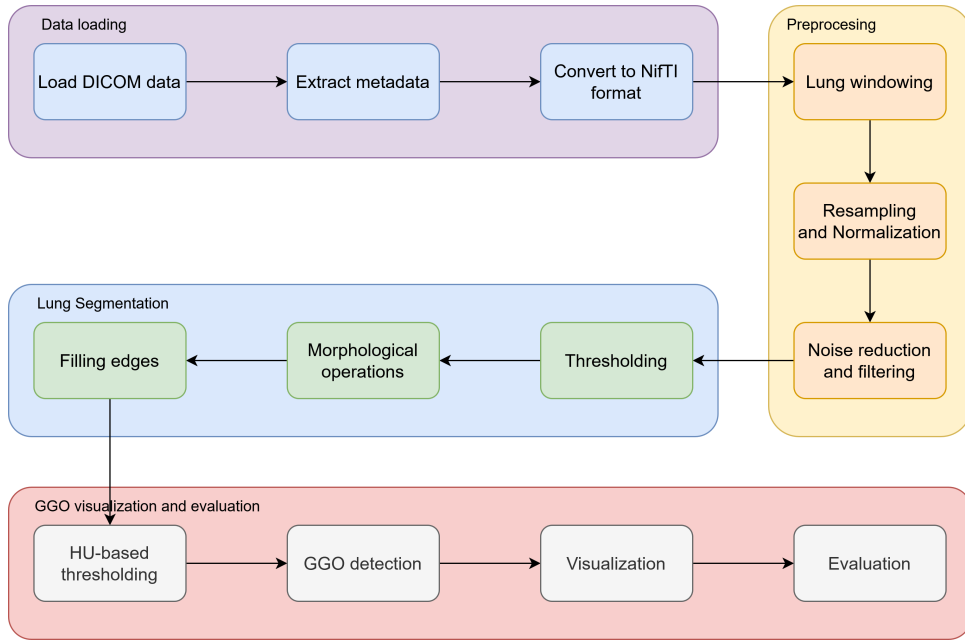


Figure 1: Methodology Overview

4.1. Data Collection and loading

We will use the publically available LIDC-IDRI dataset [7] which contains lung CT scans from the Lung Image Database Consortium (LIDC) and Image Database Resource Initiative (IDRI). The dataset contains over 11000 clinical thoracic CT scans in DICOM format. The Python library "pydicom" is the standard tool used to query and visualize this data. The data will be loaded using Python libraries such as "pydicom".

Once the dataset is loaded, we will extract their metadata. The metadata is primarily contained within the DICOM headers of the images and the XML files that hold the radiologist's findings. The LIDC-IDRI dataset provides metadata under three categories:

- **Scan and Patient information:** Includes slice thickness, pixel spacing, intra-venous contrast usage

- **Annotations:** Includes type of Lesion and ROI maps
- **Nodule characteristics:** For nodules $\geq 3\text{mm}$, the metadata includes subjective ratings from 1 to 5 for nine specific features

Once the metadata is extracted, we will convert the DICOM images to standard NiftI format using the "dcm2nii" tool for easier processing in later stages. This conversion helps for compatibility with various medical image processing libraries as DICOM is complex and bulky for analysis.

4.2. Data preprocessing

After loading and extracting metadata, the data will be preprocessed to improve the overall quality of the CT images before segmentation. This step includes:

- Windowing the images to focus on lung tissue
- Resampling and Normalizing pixel intensities
- Removing noise and artifacts

We will apply a lung-specific window level and width to enhance the visibility of lung tissues. This is a medical image convention used to clip the Hounsfield range is by choosing a central intensity. Next, resampling will be done to ensure uniform voxel spacing across all scans, which is important for accurate segmentation. This is done because, different scanners may produce images with varying resolutions. Resampling changes the resolution to a standard size such as $1\text{mm} \times 1\text{mm} \times 1\text{mm}$. We will use interpolation methods such as linear or cubic interpolation for this purpose.

Once resampling is done, normalization of pixel intensities will be performed to standardize the intensity values across different scans. This makes mathematical operations more stable and removes dependency on absolute HU values for algorithms. Noise reduction techniques such as Gaussian, Median or Bilateral filtering will be applied to reduce random variations in pixel intensities while preserving important structures like GGO boundaries. Since CT images inherently contain quantum noise, it can create false GGO detections.

4.3. Lung Segmentation

In order to perform Lung segmentation, we will use a combination of intensity based thresholding and morphological image processing techniques. First we will find the real area of the lung in mm^2 by finding the real size of the pixel dimensions. Next, we will binarize the images using intensity thresholding to separate lung tissues from the background. We expect lungs to be in the Hounsfield unit range of $[-1000, 300]$. To this end, we need to clip the image range to $[-1000, 300]$ and binarize the values to 0 and 1.

Then we need to find the contours of the lung areas. In computer vision, a contour is a set of points that describe a line or area. So for each detected contour we will not get a

full binary mask but rather a set with a multiple x and y values. In order to isolate the desired area we will use the marching squares algorithm[1]. Using the possible contours, we will extract the lungs using 2 constraints:

- The contour of the lungs must be a closed set
- The contour must have a minimum volume of 2000 pixels to represent the lungs

Using these 2 assumptions, we will convert the contours into a binary mask representing the lung areas by using the "pillow" Python library to draw a polygon connecting the contours. After obtaining the binary mask of the lungs, we will save it as a NifTI file and use it for further processing. Finally, we will do element-wise multiplication between the CT image and the lung mask we obtained to get only the lungs.

4.4. GGO Visualization and Evaluation

After segmenting the lungs, we will focus on visualizing the GGO regions within the lung areas. We will assume that if there is a pixel with an intensity value between -500 HU and -700 HU inside the lung area then we will consider it as a GGO area. However, this threshold range may also include vessels within the lung parenchyma. To address this issue, we will apply vessel removal techniques to reduce false positive detections. Since vessels appear as elongated, tubular structures, we will use morphological opening operations using a circular structuring element to remove them. Then, we will set the zeros that resulted from the previous element-wise multiplication to -1000 and finally keep only the intensities that are between -500 and -700 as GGO areas.

Once the GGO area has been identified, we will visualize it by overlaying the detected GGO regions on the original CT images using pseudo-coloring techniques. We will also visualize the original image alongside the segmented GGO area side-by-side using a User Interface. Finally, we will evaluate the effectiveness of the GGO segmentation using manually annotated contours provided in the XML files of the LIDC-IDRI dataset. [Back to Table of Contents](#)

5. Dataset / Data Collection

(Maximum word limit: 100 words) This project utilizes the LIDC-IDRI dataset obtained from The Cancer Imaging Archive (TCIA). The dataset consists of over 1,000 chest CT scans stored in DICOM format. The key statistics of the dataset are summarized as follows:

- **Total cases:** 1,018 patients
- **Total lesions:** 7,371 lesions marked as “nodule” by at least one radiologist
- **Nodules ≥ 3 mm:** 2,669 lesions identified as clinically significant nodules

Images: Clinical thoracic CT scans in DICOM format.

Annotations: Provided in XML files containing the coordinates of nodule boundaries (contours) and associated radiological characteristic ratings.

Tools: The Python library `pylidc` is used to query, process, and visualize the LIDC-IDRI dataset.

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6. Expected Outcomes

The expected outcomes of this project include:

- A Python based image processing pipeline for lung CT analysis
- A GUI application for loading, visualizing, and navigating CT slices
- Accurate lung segmentation masks
- Highlighted Ground Glass Opacity (GGO) regions with color overlays
- Quantitative outputs such as GGO count, area, and estimated volume
- Adjustable HU threshold controls for experimentation
- Side by side visualization of original and processed images

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7. Percentage of Image Processing & Computer Vision Involvement

(Approximately 50–100 words) Core components include DICOM loading, Hounsfield Unit extraction, lung windowing, threshold-based lung segmentation, morphological operations, connected component analysis, flood fill, and GGO detection using HU ranges. Vessel suppression is performed using structural morphology and region-based filtering. Dataset preprocessing and visualization parts of the pipeline are the only components that do not involve image processing or computer vision techniques.

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- [2] J. W. Song, J. G. Im, M. I. Ahn, *et al.*, “Ct findings for ground-glass attenuation in subacute hp,” *Journal of Thoracic Imaging*, vol. 16, no. 3, pp. 234–242, 2001.
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- [7] S. G. Armato III, G. McLennan, L. Bidaut, M. F. McNitt-Gray, C. R. Meyer, A. P. Reeves, B. Zhao, D. R. Aberle, C. I. Henschke, E. A. Hoffman, E. A. Kazerooni, H. MacMahon, E. J. R. Van Beek, D. Yankelevitz, A. M. Biancardi, P. H. Bland, M. S. Brown, R. M. Engelmann, G. E. Laderach, D. Max, R. C. Pais, D. P.-Y. Qing, K. J. Roberts, B. R. Smith, A. Starkey, P. Batra, P. Caligiuri, A. Faridzadeh, G. W. Gladish, C. M. Jude, R. F. Munden, I. Petkovska, L. E. Quint, L. H. Schwartz, B. Sundaram, L. E. Dodd, C. Fenimore, D. Gur, N. Petrick, J. Freymann, J. Kirby, S. Moore, R. Paden, and R. Sethi, “The lung image database consortium (lidc) and image database resource initiative (idri): A completed reference database of lung nodules on ct scans,” *Medical Physics*, vol. 38, no. 2, pp. 915–931, 2011.

A minimum of ten (10) references must be included in the References section.